V SEMESTER B.Tech. M&C
COURSE PLAN: THEORY COURSE

School Name	**Computer Engineering**				
Course Name	**Fundamentals of Machine Learning**				
Course Code	**CSS 3131**	**Semester**	**V**	**Core / PE / OE**	**Core**
Name of the faculty	**Dr. Mamatha Balachandra**				

No of contact hours/week:	**L**	**T**	**P**	**C**
	3	**0**	**1**	**4**

Mapping of Course Outcomes (COs)/Course Learning Outcomes (CLOs) #

CO/ CLO	**Statements**	**No. of Contact Hours**	**Marks**	**Program Outcomes (POs)**	**Learning Outcomes #**	**BL* *(Recommended)***
1	Apply appropriate Machine Learning techniques, including K-Nearest Neighbors, Linear Regression, and Logistic Regression, to solve basic prediction and classification problems.	17	24 M	1,2,5 &11	1,3	2 to 6
2	Analyze supervised learning algorithms like logistic regression, k-nearest neighbors, and Naïve Bayes for classification and regression, considering bias-variance trade-offs and cost functions.	25	36 M	1,2,5 &11	3,5	2 to 6
3	Evaluate advanced Supervised learning techniques, including decision trees, support vector machines, and artificial neural networks, with a focus on their implementation, optimization, and applicability.	12	16 M	1,2,5 &11	2	2 to 6
4	Apply unsupervised learning methods such as clustering and dimensionality reduction, along with ensemble techniques like boosting, bagging, and stacking, to address challenges like anomaly detection and class imbalance.	10	14 M	1,2,5 &11	2,3	2 to 6

5	Apply basic Artificial Neural Network techniques and dimensionality reduction methods to solve machine learning problems involving high-dimensional data.	8	10 M	1,2,5 &11	1,3,5	2 to 6
	Total	**72**	**100 M**			

Applicable to IET Accredited Programs Only

*******Preferred BL levels are above 3 (apply)*

Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3	PSO4
CO1	3	2									2	3	2	3	
CO2	3	2	3		2						2	3	3	3	
CO3	3	2	3		2						2	3	3	3	
CO4	3	2	3	2	2						2	3	3	3	
CO5	3	2	3	2	2						2	3	3	3	
Average Articulation Level	**2.5**	**1.67**	**2**	**0.6**	**1.3**	**0**	**0**	**0**	**0**	**0**	**1.67**	**2.5**	**2.5**	**2.5**	

Module Matrix#

	LO1	LO2	LO3	LO4	LO5	LO6	LO7	LO8	LO9	LO10	LO11	LO12	LO13	LO14	LO15	LO16	LO17	LO18
CLO1	**B**		**B**															
CLO2			**B**		**B**													
CLO3		**B**																
CLO4		**B**	**B**															
CLO5	**B**		**B**		**B**													
CLO6																		

Applicable to IET Accredited Programs Only. Fill with B or C

Delivery and Assessment Plan of LOs#

Learning Outcome (LO) mapped to the course		**Delivery and assessment Plan**
LO	**LO statement**	
01	Apply appropriate Machine Learning techniques, including K-Nearest Neighbors, Linear Regression, and Logistic Regression, to solve basic prediction and classification problems.	PPT and Case studies
02	Analyze supervised learning algorithms like logistic regression, k-nearest	PPT and Case studies

	neighbors, and Naïve Bayes for classification and regression, considering bias-variance trade-offs and cost functions.	
03	Evaluate advanced Supervised learning techniques, including decision trees, support vector machines, and artificial neural networks, with a focus on their implementation, optimization, and applicability.	PPT and Case studies
04	Apply unsupervised learning methods such as clustering and dimensionality reduction, along with ensemble techniques like boosting, bagging, and stacking, to address challenges like anomaly detection and class imbalance.	PPT and Case studies
05	Apply basic Artificial Neural Network techniques and dimensionality reduction methods to solve machine learning problems involving high-dimensional data	PPT and Case studies

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The Assessment Plan shall be formulated as per the directives communicated by the Office of the AD(A) in every odd semester.

Evaluation plan- Integrated (Theory + Lab)

MANIPAL
ACADEMY of HIGHER EDUCATION
(Deemed to be University under Section 3 of the UGC Act, 1956)

ANNEXURE 3

Minutes of 1st BoS Meeting: School of Basic Sciences, Humanities and Management

Date & Time: Apr 27, 2026, 10.30am *Venue*: *Senate Hall, MAHE Manipal*

Attendance: Sheet Attached

Agenda points:

1. Assessment plan for Lab integrated theory courses, applicable from AY 2026-27
2. Exit policy for the Dual degree students, applicable from AY 2026-27
3. Revision to the publication criteria for PG/BTech honours, applicable from AY 2026-27
4. Approval for percentile-based project grading policy, applicable from AY 2026-27
5. Revision of syllabus and assessment pattern of mandatory PhD coursework:Advanced Research Methodology, applicable from AY 2026-27
6. Mandatory Academic Project completion for all BTech Students, applicable from AY 2026-27
7. Approval for new open electives for BTech students of MIT Manipal, applicable from AY 2026-27
8. Approval for Swayam equivalent course, effective from AY 2025-26
9. Approval for new Open Elective courses offered by Science, Humanities and Management Department, MIT Bengaluru, applicable from AY 2026-27
10. Agenda for academic council
11. Any other matter

Sl. No.	Agenda Point	Discussion	Action	Responsibility	Target date
1	Assessment plan for Lab integrated theory courses, applicable from AY 2026-27	• Laboratory work evaluated out of 100 marks • Final lab marks shall be scaled down to 20 marks • Total internal marks shall comprise: o Mid-term Exam (Theory): 30 marks (may cover a lab question) o Lab continuous Assessments: 20 marks o Total Internal: 50 marks • End-Semester Examination (Theory): 50 Marks • There shall be no separate grade or failure status ("F") for the laboratory component. These guidelines shall be applicable to all lab-integrated theory courses offered by the Institute with immediate effect. (Refer Annexure 1). Registrar Evaluation has expressed that we can try this assessment plan for a year and correct it later, if necessary, based on the feedback from students and faculty.	Discussed and approved	—	Applicable from AY 2026-27.

Annexure 1

Guidelines for Assessment and Evaluation of Lab-Integrated Theory Courses- (Applicable from AY 2026-27)

Purpose

This guidelines defines the assessment structure, evaluation process, and grading norms for lab-integrated theory courses to ensure uniformity, transparency, and alignment with credit-based weightages.

1. Course Structure and Credit Weightage

- Lab-integrated theory courses comprise:
  - Theory: 3 Credits
  - Laboratory: 1 Credit
- Assessment weightage shall reflect the credit ratio 3:1, equivalent to:
  - Theory: 75%
  - Lab: 25%

2. Marks Distribution

A. Theory Component (Total: 75 Marks)

- Mid-term Examination: 25 marks + 5 marks(lab specific questions)
- End-Semester Examination: 50 Marks(may include lab questions)

B. Laboratory Component (Total: 20 Marks)

- Laboratory work evaluated out of 100 marks
- Final lab marks shall be scaled down to 20 marks
- Labs conducted as regular laboratory sessions (e.g., 08-12 experiments)

3. Assessment Workflow

- Mid-term examinations and end-semester examinations shall be conducted through ePAD.
- Laboratory assessment shall be continuous and conducted during scheduled lab sessions.
- Total internal marks shall comprise:
  - Mid-term Exam: 30 marks
  - Lab continuous Assessments: 20 marks
  - Total Internal: 50 marks
- Overall course evaluation shall be out of 100 marks (Internal + External).

4. Evaluation and Grading

- Evaluation and grading shall follow the same norms as UG theory courses.
- There shall be no separate grade or failure status ("F") for the laboratory component.

5. Attendance and Failure Policy

- Laboratory attendance requirements shall be identical to those of regular laboratory courses.
- If a student fails in a lab-integrated theory course, the entire course including the laboratory component must be repeated.

This guidelines shall be applicable to all lab-integrated theory courses offered by the Institute with immediate effect.

Evaluation Plan: ML LAB

- Internal Assessment Marks: 60M
  - Continuous Evaluation: 20M
    - Continuous evaluation component (for each evaluation)
    - The assessment will depend on punctuality, program execution, maintaining the observation note and answering the questions in viva voce.
  - Internal-Assessment : 20M

No of Assignments	Marks	Mode
3 assessments	10+5+5	• 10 – Surprise test-Pen and Paper • 5+5 - Course Coordinator can decide

Schedule for Internal Assessment (IA) Components (Session: July – Nov 2026)

(For Courses Opting to Conduct 3 Components Only)

Component Number	Schedule / Date Range
IA1	17 Aug 2026 – 29 Aug 2026
IA2	7 Sep 2026 – 21 Sep 2026
IA3	19 Oct 2026 – 31 Oct 2026

Note:

- IA1 will be conducted in the designated classroom(s) from 4:40 PM to 5:20 PM on the aforementioned date(s).
- IA2 and IA3 will be conducted through the LMS from 5:30 PM to 5:50 PM on the aforementioned date(s).
- Mini-project: 20M [Report 50% + Implementation and Demo 50%]

- End-of-semester assessment: 40 M

  Note: Scheduled in between Oct. 25- Nov 10, 2026, and the syllabus is Week 02 through Week-08.

Lesson Plan (Theory)

L No	Topics	CO Addressed
0	*To address OBE philosophy towards NBA and IET accreditation followed by a discussion on Course Plan, Course Outcomes, CO-PO Mapping and Blooms Taxonomy, with prerequisites for the course, if any.*	
1	Introduction to Machine Learning, Classification of ML algorithms.	CO 1
2	Statistical Learning: What is Statistical Learning?	CO 1
3	Parametric Methods and Non- Parametric Models	CO 1
4	Supervised Versus Unsupervised Learning, Regression Versus Classification Problems	CO 1
5	The Trade-Off Between Prediction Accuracy and Model Interpretability, Assessing Model Accuracy, Measuring the Quality of Fit, The Bias-Variance Trade-Of	CO 1
6	Linear Regression: *Simple Linear Regression, Estimating the Coefficients, Problems on Simple Linear Regression	CO 1
7	Problem on Simple Linear Regression, Assessing the Accuracy of the Model	CO 1
8	*Normal Method, *Gradient Descent Method, *Normal Equation Method vs Gradient Descent Method	CO 1
9	MSE, RMSE, R^2 Statistic, *Multiple Linear Regression, Some Important Questions, Problems on Multiple Linear Regression	CO 1
10	Polynomial Regression, Problems on Polynomial Regression	CO 1
11	Classification: An Overview of Classification, Why Not Linear Regression?, *Logistic Regression, Sigmoid Function, *Problems on Logistic Regression	CO 2
12	The Logistic Model, Estimating the Regression Coefficients, Making Predictions, Multinomial Logistic Regression	CO 2
13	*Cost Function, * STOCHASTIC GRADIENT DESCENT on Logistic Regression	CO 2
14	*Performance metrics: Confusion Matrix, Precision and Recall & ROC curve	CO 2
15	*BAYESIAN REASONING: *Bayes Theorem and Example Problem on Bayes theorem	CO 2
16	Working of Naïve Bayes Classifier, Key Concepts of Naïve Bayes Classifier	CO 2
17	Naïve Bayes with multiple features - Example	CO 2
18	Working of Naïve Bayes Classifier –Text Classification	CO 2
19	*K-Nearest Neighbour(kNN) Classifier, Comparison of Linear Regression with K-Nearest Neighbors	CO 2
20	Applications of kNN, working of k-NEAREST NEIGHBOR (k-NN) CLASSIFIER	CO 2
21	Problem on kNN for classification and regression	CO 2
22	*Need for feature scaling	CO 2
23	*Decision Trees: Introduction, Example of a Classification Decision Tree, Measure of Impurity for Evaluating Splits in Decision Trees	CO 3
24	*ID3, *C4.5, *CART Decision Trees	CO 3
25	*Pruning the Tree, *Strengths and Weaknesses of Decision Tree Approach	CO 3
26	Unsupervised Learning: The Challenge of Unsupervised Learning, Principal Components Analysis	CO 5
27	Applications of PCA	CO 5
28	Problems on PCA	CO 5
29	Missing Values and Matrix Completion	CO 5

29	*Clustering Methods, *K-Means Clustering, *Problems	CO 4
30	*Hierarchical Clustering, *Problems, Practical Issues in Clustering	CO 4
31	Bagging & Boosting algorithms	CO 4
32	To address challenges like anomaly detection and class imbalance	CO 4
33	Ensembling and Boosting Algorithms: Concept of weak learners	CO 4
34	Support Vector Machines: Maximal Margin Classifier- What Is a Hyperplane?, Classification Using a Separating Hyperplane	CO 3
35	Support Vector Classifiers-Overview of the Support Vector Classifier, Details of the Support Vector Classifie, *Numerical examples	CO 3
36	Support Vector Machines-Classification with Non-Linear Decision Boundaries, The Support Vector Machine, An Application to the Heart Disease Data, *Numerical examples	CO 3

***Refer the PPT**
-(LAB) No instructions were given during the class.

Lesson Plan (ML Lab)

L No	Topics	CO Addressed
1	Week-01: Working with NumPy, SciPy, and CSV file format	CO 1
2	Week-02: Simple Linear Regression with SGD	CO 1
3	Week-03: Multiple and Polynomial Linear Regression	CO 1
4	Week-04: Logistic Linear Regression with SGD	CO 2
5	Week 05: Cost functions	CO 2
6	Week-06: K-Nearest Neighbor (k-NN)	CO 2
7	Week-07: Naïve Bayes Classifier	CO 2
8	Week-08: Decision Tree	CO 2
9	Week-09: Mini-project	CO 3, 4 & 5
10	Week-10: Mini-project Phase I	CO 3, 4 & 5
11	Week-11: Mini-project	CO 3, 4 & 5
12	Week-12: Mini-project Phase II	CO 3, 4 & 5

Self Directed Learning (SDL)

Sl. No.	TOPIC	No. of learning hours	CO
1	Machine Learning With Python [Machine Learning and Deep Learning With Python: Use Python Jupyter to Implement Mathematical Concepts, Machine Learning Algorithms and Deep Learning Neural Networks - Search]	08	CO 1
2	Machine Learning Foundations Course – Regression Analysis [Bing Videos]	06	CO 2

3	Stochastic Gradient Descent [https://www.ibm.com/think/topics/stochastic-gradient-descent] (REF Text Book 02)	04	CO 2
4	Generative Models for Classification: Linear Discriminant Analysis for p =1, Linear Discriminant Analysis for p>1, Quadratic Discriminant Analysis, Naive Bayes) (REF Text Book 01)	04	CO 2
5	Principal Component Analysis (REF Text Book 02)	04	CO 2
6	Naive Bayes Case Studies-Healthcare: Disease Diagnosis using Naive Bayes, Cardiovascular Disease Diagnosis using Improved Naive Bayes[https://www.researchsquare.com/article/rs-1231978/latest.pdf]; Sports: Football Match Result Prediction using Naive Bayes [https://medium.com/@simon.tettmar/data-science-ca87a98d5637]	04	CO 2
7	K-Nearest Neighbors (K-NN) Case Studies-Healthcare: Disease Risk Prediction using KNN [https://www.nature.com/articles/s41598-022-10358-x], Precision Healthcare Diagnosis using KNN[https://sites.psu.edu/mcreu/2024/07/22/precision-healthcare-diagnosis-through-advanced-predictive-modelling-harnessing-the-power-of-k-nearest-neighbour-algorithm/], Predicting Medical Conditions using KNN [https://bpb-us-e1.wpmucdn.com/wordpressua.uark.edu/dist/d/728/files/2017/07/2.pdf]	04	CO 2
7	Decision Tree case studies and examples Healthcare Case Studies: Adverse Drug Reaction Classification using Decision Trees [https://medium.com/thedeephub/data-science-project-build-a-decision-tree-model-with-healthcare-data-fe93d2d6c70c], Healthcare Decision Tree Examples [https://betanet.net/view-post/decision-tree-examples-in-healthcare-a]; Sports Case Studies: Football Analytics using Decision Trees [https://blog.gopenai.com/enhancing-football-analytics-with-decision-trees-a-case-study-on-passing-and-interception-analysis-9f3821b4da89], Sports Training Data Analysis using ID3 Decision Tree [https://www.nature.com/articles/s41598-025-99996-5], Sports Analytics Case Studies [https://www.samford.edu/sports-analytics/case-studies]	04	CO 3
8	Support Vector Machine (SVM) case studies-Public Health Analytics using SVM [https://pmc.ncbi.nlm.nih.gov/articles/PMC11806959/], Real-Time Healthcare Monitoring using SVM [https://jisis.org/wp-content/uploads/2026/03/2026.I1.048.pdf], Medical Diagnosis using Multi-Class SVM [https://www.mdpi.com/1424-8220/23/22/9033]; Sports Analytics SVM Case Studies: Athlete Injury Classification using SVM [https://ieeexplore.ieee.org/document/10317743], Sports Analytics using SVM [https://github.com/nani67/SportsAnalytics_ML_DL]	04	CO 4
9	Clustering (K-Means, Hierarchical Clustering, DBSCAN) case studies-Healthcare Clustering Case Studies: Patient Segmentation using Clustering Techniques, Patient Clustering Optimization with K-Means [https://ieeexplore.ieee.org/document/10921818]; Sports Clustering Case Studies: Clustering Football Teams using K-Means [https://medium.com/@javiermazario9/clustering-of-football-teams-using-k-means-english-version-bc433267e908], Kabaddi Performance Analysis using K-Means [https://link.springer.com/article/10.1007/s11042-023-16819-3]	04	CO 5

10	COMBINING MULTIPLE LEARNERS: Rationale, Generating Diverse Learners, Voting, Error-Correcting Output Codes, Bagging, Boosting [https://scikit-learn.org/stable/modules/ensemble.html; https://www.statlearning.com/]	02	CO 5
	Total Number of learning hours	**48**	

***The SDL topics may be from the Research Literature/ Case Studies NPTEL/Coursera/any MOOC-based material Virtual Lab, etc.**

****One credit is equivalent to 30 hours of learning. (4 credits course =120 hrs learning ; 48+72=120 hrs of learning)**

Faculty members teaching the course (if multiple sections exist): NA

References:

Textbooks	1. Kevin P. Murphy, "Probabilistic Machine Learning: An Introduction (Adaptive Computation and Machine Learning series)", MIT Press, 2022
	2. G. James, D. Witten, T Hastie, R. Tibshirani, An Introduction to Statistical Learning with applications in R, Springer, 2013
	3. T. M. Mitchell, Machine Learning, (Indian Edition), McGraw-Hill, 2017.
	4. C. Bishop, Neural Networks for Pattern Recognition, Oxford University Press, 2019
	5. Parteek Bhatia, Machine Learning with Python Principles and Practical Techniques, Cambridge University Press, 2024.
	6. Hastie, R. Tibshirani, J. Friedman, The Elements of Statistical Learning, (2e), Springer,2009.
Self-Directed Learning	Listed above
Research Literature/ Case Studies	Healthcare and Sports
NPTEL/Coursera/any MOOC-based material	https://www.youtube.com/playlist?list=PLqRJbI_zl_SgQW4PwXufINCHl_qwyowRB / https://nptel.ac.in/courses/106106139
Virtual Labs	Nil

Submitted by: Dr. Mamatha Balachandra

(Signature of the faculty)
Date: 10/08/2026
Approved by:

(Signature of the Dean)